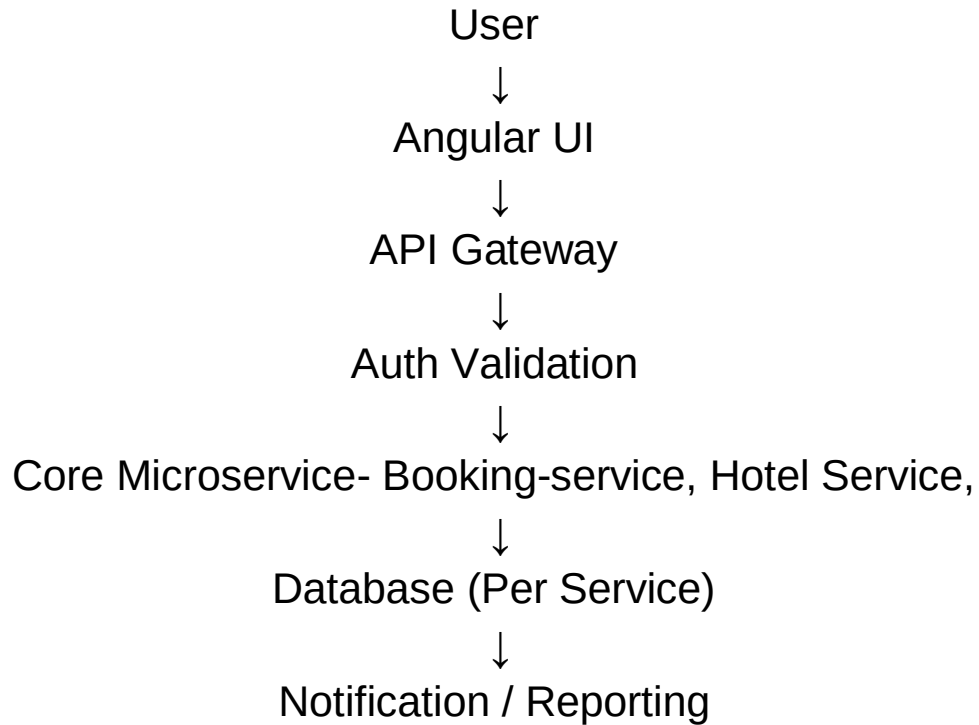


Hotel Management System

Overall Flow:-



Microservices :-

- Eureka-service
- API-Gateway
- Config-server
- Auth-service
- Booking-service
- Hotel-service
- Notification-service

Er Design

A.USER

- id (PK)
- email (unique)
- password
- hotelId (for Admin and Manager; null for Guest)
- role (ADMIN / MANAGER / RECEPTIONIST / GUEST)

B.HOTEL

- id (PK)
- hotelName
- city
- address
- numberOfRooms
- booked (bookedRooms)
- status (Active/NotActive)

C.ROOM

- id (PK)
- roomNumber
- hotelId (FK)
- status (AVAILABLE / OCCUPIED / MAINTENANCE)
- type (Single/Double/Deluxe/Suite)
- Price

D.RESERVATION

- id (PK)
- hotelId (FK)
- roomId (FK)
- guestName
- guestEmail
- checkInDate
- checkOutDate
- price
- hotelName
- status (BOOKED / CONFIRMED / CHECKED_IN / CHECKED_OUT / CANCELLED)
- checkInReminderSent
- checkOutReminderSent

Role Functionality

A. Admin

- Add hotel locations
- Delete hotel locations
- Update hotel details
- Retrieve all hotels

B. Manager

- Update hotel prices/details
- Add/delete hotel rooms
- Retrieve all rooms in a hotel
- Retrieve all bookings for a hotel

C. Receptionist

- Check in/check out
- Book hotel
- Billing

D. Guest

- Book hotels
- Cancel booking
- Update booking
- View own bookings
- Search hotels
- View available rooms
- Register and login
- Change password

Endpoints

Endpoint	Payload	Access	Function
POST /auth/register	{"email":"user@example.com","password":"password123","confirmPassword":"password123","role":"USER"}	Anyone	Register a new user account
POST /auth/login	{"email":"user@example.com","password":"password123"}	Anyone	Authenticate user and return JWT token
PUT /auth/changePassword	{"email":"user@example.com","oldPassword":"oldpass","newPassword":"newpass"}	Authenticated Guest	Change user password
POST /api/booking/{hotelId}	{"roomId":"room123","guestName":"John Doe","guestEmail":"john@example.com","checkInDate":"2024-01-15","checkOutDate":"2024-01-17"}	Guest / Receptionist	Create a new room booking
DELETE /api/booking/{email}/{reservationId}/cancel	N/A	Guest	Cancel an existing booking
GET /api/booking/{email}/all	N/A	Guest	Retrieve all bookings for a user
GET /api/booking/booking/{hotelId}	N/A	Manager / Admin	Retrieve all bookings for a specific hotel
PUT /api/booking/{email}/{reservationId}/update	{"checkInDate":"2024-01-16","checkOutDate":"2024-01-18"}	Guest	Update booking dates
PUT /api/booking/checkin/{bookingId}	{"checkIn":true}	Receptionist	Perform check-in or check-out
POST /hotel	{"hotelName":"Grand Hotel","city":"New York","address":"123 Main St","numberOfRooms":100}	Admin	Create a new hotel
PUT /hotel/{hotelId}	{"hotelName":"Updated Hotel","city":"New York","address":"456 Elm St","numberOfRooms":120}	Admin / Manager	Update hotel details
DELETE /hotel/{hotelId}	N/A	Admin	Delete a hotel
GET /hotel/all	N/A	Admin	Retrieve all hotels

Endpoint	Payload	Access	Function
POST /hotel/{hotelId}/room	{ "roomNumber": "101", "status": "AVAILABLE", "type": "SINGLE", "price": 100.0 }	Manager	Add multiple rooms to a hotel
PUT /hotel/{hotelId}/room/{roomId}	{ "status": "OCCUPIED", "type": "DOUBLE", "price": 150.0 }	Manager	Update room details
POST /search	{ "city": "New York", "checkIn": "2024-01-15", "checkOut": "2024-01-17", "roomCount": 2 }	Public	Search available hotels
POST /hotel/{hotelId}/rooms/available	{ "checkIn": "2024-01-15", "checkOut": "2024-01-17" }	Public	Get available rooms for given dates
GET /rooms/{hotelId}	N/A	Manager	Retrieve all rooms in a hotel
PUT /{hotelId}/rooms/{roomId}/{bookingId}	{ "checkIn": true }	Receptionist	Check-in or check-out for a room
DELETE /hotel/{hotelId}/room/{roomId}	N/A	Manager	Delete a room from a hotel

Business rules

A.User & Security Rules

1. Every user must authenticate using **JWT-based authentication**
2. Role-Based Access Control (RBAC) is enforced at API level
3. Passwords are stored using strong encryption (BCrypt)
4. Guests can access only permitted APIs based on role
5. Tokens are required for all protected endpoints

B.Hotel & Room Rules

1. A hotel can have multiple room categories
2. A room category defines:
 - Maximum occupancy
 - Base price
3. Each physical room belongs to:
 - One hotel
 - One room category
4. Room status must be one of:
 - Available
 - Occupied
 - Maintenance
 - Rooms under Maintenance cannot be booked

C.Reservation must include:

- Guest
- Hotel
- Room category
- Check-in date
- Check-out date

D. Reservation rules:-

1. A room **must not be double-booked** for overlapping dates
2. Reservation modification is allowed only before check-in
3. Cancelled reservations release room availability.

F. Check-in / Check-out Rules

1. Check-in:
 - Allowed only for Confirmed reservations
 - Room status becomes Occupied
2. Check-out:
 - Ends reservation
 - Room status becomes Available
3. Walk-in bookings are allowed by Receptionist only

Architecture

